CS 585 Term Project Final Presentation: Integrating Bidirectional Mamba into HiFi-GAN for Long-Range Speech Synthesis

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Audio Demo: Our model says hello!

▶ **Play: "Hello, we are presenting our project"**

System Overview

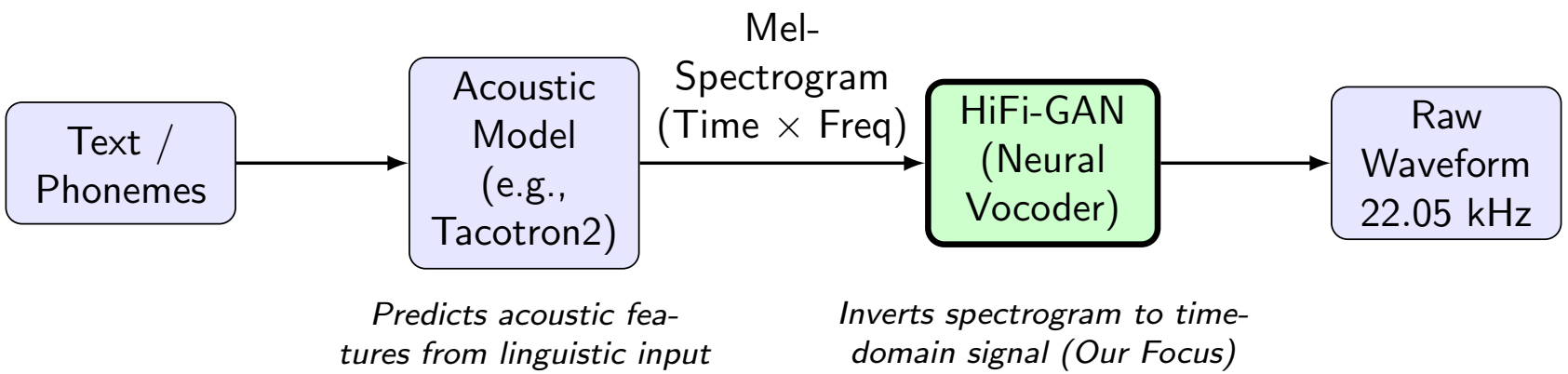


Figure: Text-to-Speech pipeline: Our work focuses on the neural vocoding stage

Training Objective

Given mel-spectrogram $M \in \mathbb{R}^{F \times L}$, learn generator G_θ to synthesize waveform $\hat{S} = G_\theta(M)$

$$\mathcal{L}_G = \underbrace{\mathbb{E}[(D(G(M)) - 1)^2]}_{\text{Adversarial Loss}} + \underbrace{\lambda_{\text{fm}} \sum_{i=1}^{T} \mathbb{E}\left[\|D_i(S) - D_i(G(M))\|_1\right]}_{\text{Feature Matching Loss}} + \lambda_{\text{mel}} \underbrace{\|\text{Mel}(S) - \text{Mel}(G(M))\|_1}_{\text{Mel-Spectrogram Loss}}$$

Motivation: Why Improve HiFi-GAN?

Problem: Neural vocoders must model two competing objectives:

Local Texture

High-frequency details

- Timbre, fricatives
- Transient attacks
- Periodic patterns

✓ CNNs excel at this

Global Structure

Long-range dependencies

- Prosody, pitch contours
- Speaker identity
- Sentence-level coherence

× CNNs have limited receptive field

Goal: Combine CNN's local modeling with SSM's global context

Baseline: HiFi-GAN Architecture

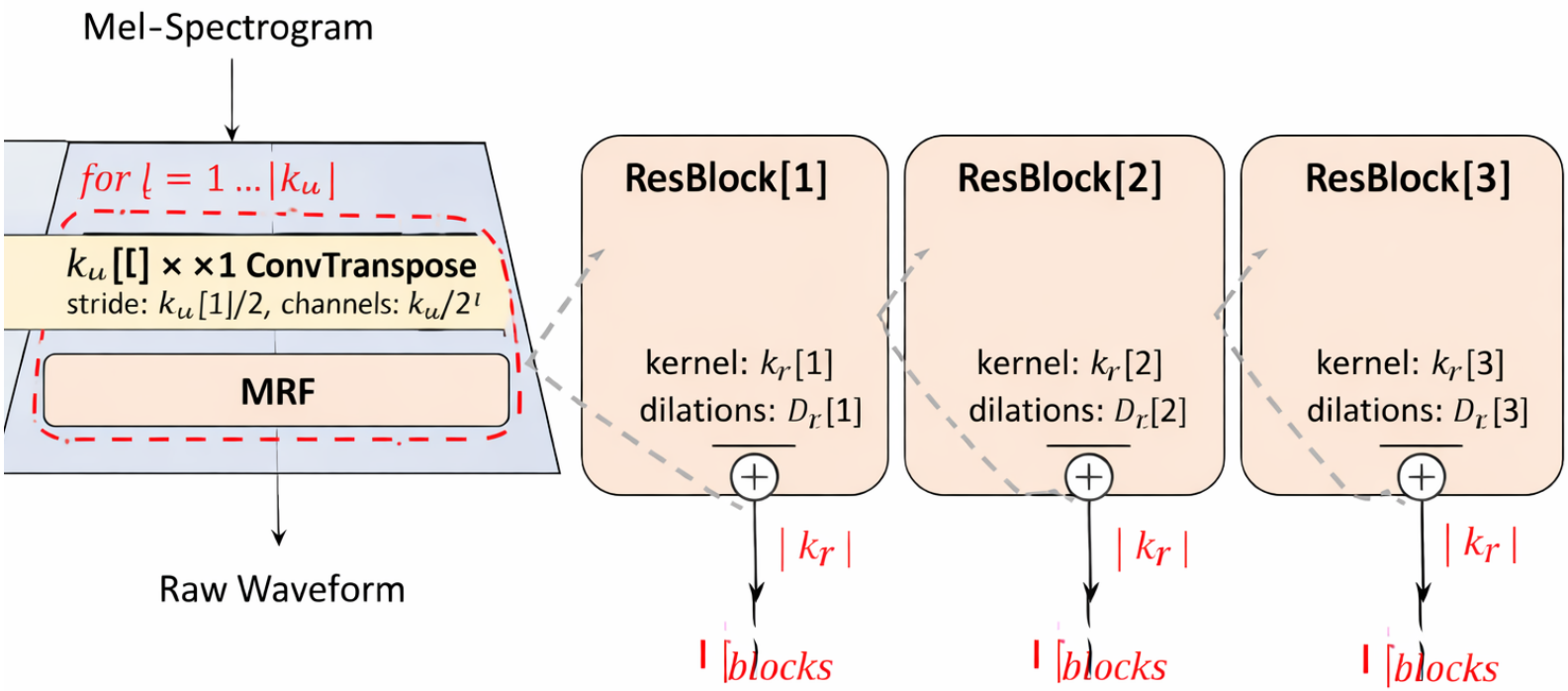


Figure: HiFi-GAN generator with Multi-Receptive Field (MRF) fusion module

Strength:

- Fast inference

Limitation:

- Finite receptive field

The Receptive Field Bottleneck

Mathematical Constraint

A convolutional layer with kernel size k and dilation d has receptive field:

$$\text{RF} = k + (k-1) \cdot d$$

Even with deep stacking, **RF remains bounded**

Consequence

Output sample $\hat{s}_t$ depends only on local window $[t-k, t+k]$

Prosodic features span thousands of samples (far beyond typical RF)

Solution: State Space Models (SSMs) $\rightarrow$ Theoretically infinite context with $\mathcal{O}(L)$ complexity

Proposed Solution: Hybrid Architecture

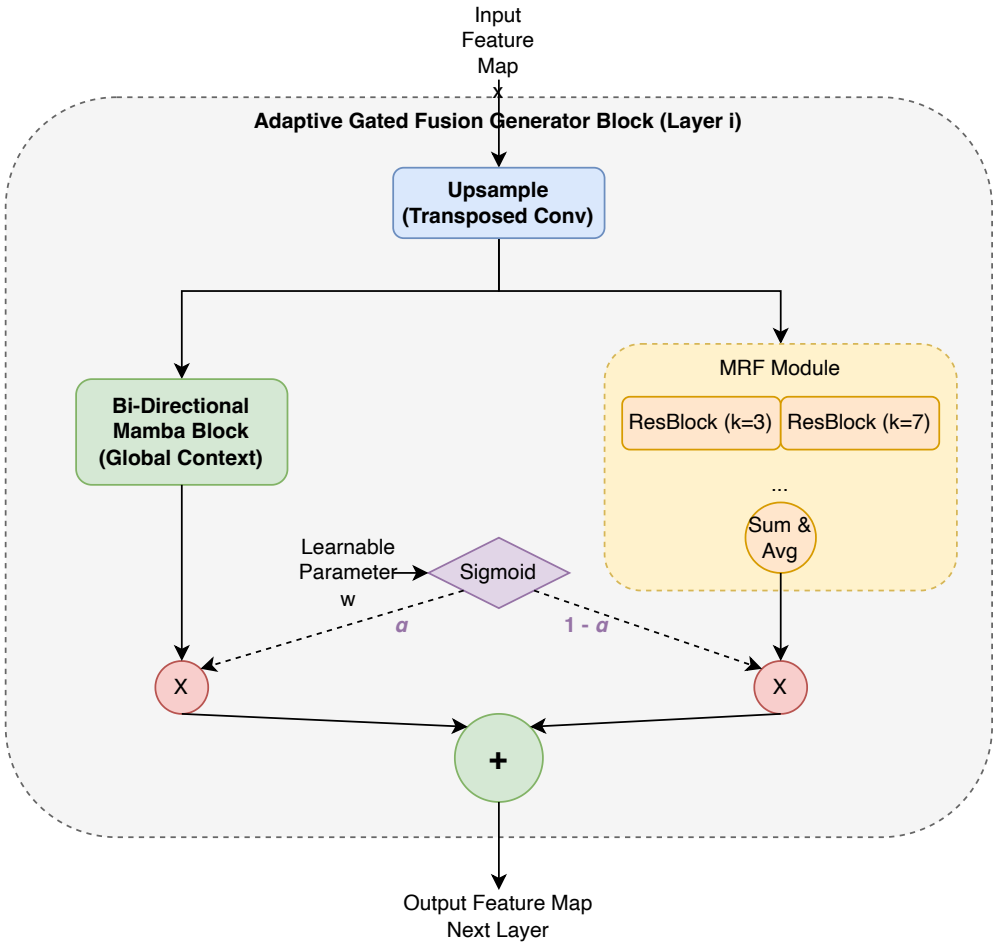


Figure: Parallel integration of Bidirectional Mamba and MRF with Adaptive Gated Fusion

Architecture Components

1. Bidirectional Mamba

Why bidirectional?

- Vocoding is non-causal
- Need past & future context

Implementation:

$$Y_{\text{fwd}} = \text{SSM}(X)$$
$$Y_{\text{bwd}} = \text{Flip}(\text{SSM}(\text{Flip}(X)))$$
$$\Phi = \text{LN}(Y_{\text{fwd}} + Y_{\text{bwd}})$$

2. Gated Fusion

Why gating?

- Static averaging fails
- Resolution-dependent balance needed

Behavior:

- Early layers (low-res): High $\alpha \rightarrow$ Emphasize Mamba for prosody
- Late layers (high-res): Low $\alpha \rightarrow$ Emphasize CNN for texture

Key Insight: The network *learns* when to rely on global vs. local features

Experimental Setup

Dataset

LJSpeech

- 24 hours, single speaker
- 13,100 clips @ 22.05 kHz
- Clean, studio-quality

Evaluation Metrics

- **Feature Matching Loss**
  (Perceptual quality)
- **L1 Mel-Spec Loss**
  (Spectral fidelity)

Compared Architectures

- **Baseline:** Standard HiFi-GAN (Kong et al., 2020)
- **Ablation 1:** Mamba Only (no convolutions)
- **Ablation 2:** Hybrid with naive summation
- **Proposed:** Hybrid with Adaptive Gated Fusion (Mamba64 & Mamba128)

Reference: Kong et al., "HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis," NeurIPS 2020.

Quantitative Results

Model Configuration	FM Loss ↓	L1 Spec Loss ↓
Baseline (HiFi-GAN)	0.0736	0.2975
Ablation 1: Mamba Only	0.1291	0.3307
Ablation 2: Hybrid Summation	0.0711	0.2991
Hybrid 1 Gated (Mamba64)	**0.0623**	0.2806
Hybrid 1 Gated (Mamba128)	0.0771	**0.2587**

Table: Performance on LJSpeech validation set. Lower is better.

Perceptual Quality

15.3%

improvement in FM Loss
(Mamba64 Gated)

Spectral Fidelity

13.0%

improvement in L1 Loss
(Mamba128 Gated)

Ablation Study: Why Gating Works

Key Finding: Each design choice serves a specific purpose

Configuration	Failure Mode	FM Loss
Baseline	—	0.0736
Mamba Only	Lacks local texture	0.1291
Hybrid Summation	Static summation insufficient	0.0711
Gated Fusion	✓ Dynamic balance	**0.0623**

Interpretation

- Pure Mamba **cannot** synthesize high-frequency texture (needs CNN inductive bias)
- Simple averaging forces compromise at all resolutions
- **Learnable gating** allows layer-wise specialization:
  - Low-res → Favor Mamba (structure)
  - High-res → Favor CNN (texture)

Systematic Exploration: What Didn't Work

We tested **12 alternative integration strategies**. Failures clustered into 3 categories:

Phase Issues

Examples:

- Post-Net Refinement
- Dual-Path Decoupled
- Multi-Rate Hybrid

Cause:
Destructive interference, aliasing artifacts

Gradient Issues

Examples:

- FiLM Modulation
- Sequential Progressive
- Mamba-SE

Cause:
Unstable training, gradient vanishing

Bottleneck Issues

Examples:

- Concatenation Fusion
- Global Broadcasting
- Pre-Net Refinement

Cause:
Information loss, interpolation artifacts

Lesson: Success requires matched-resolution parallel processing with learnable fusion

Discussion: Trade-offs and Insights

Computational Cost

Inference Time Overhead:

- Mamba64: $\approx +10\%$ vs. baseline

Still **real-time capable**
(Linear $\mathcal{O}(L)$ complexity)

Metric Divergence

- Mamba64: Best *perceptual*
- Mamba128: Best *spectral*

Implication:
Discriminator-driven FM Loss prioritizes artifact suppression
L1 Loss captures prosodic accuracy

Key Takeaways

1. Global context is **not** a replacement for local texture—it's a complement
2. The optimal balance is **resolution-dependent** (gating discovers this automatically)
3. Phase coherence is **fragile**—any post-processing or multi-rate design risks artifacts

Qualitative Evaluation: Audio Samples

Listen to the synthesized audio samples:

Model	Audio Sample
Ground Truth	▶ Play
Baseline HiFi-GAN	▶ Play
Hybrid 1 Gated (Mamba64)	▶ Play
Mamba Only (Failed)	▶ Play

Listening Instructions

Focus on:

- **Prosodic smoothness** (pitch contour consistency)
- **Fricative clarity** (high-frequency texture)
- **Naturalness** (absence of metallic artifacts)

Note: Audio files should be placed in ./audio/ directory relative to this .tex file

Conclusion

Summary:

- **Problem:** HiFi-GAN's finite receptive field limits global prosodic modeling
- **Solution:** Hybrid architecture with Bidirectional Mamba + Adaptive Gated Fusion
- **Results:**
  - 15.3% improvement in perceptual quality (FM Loss)
  - 13.0% improvement in spectral fidelity (L1 Loss)
- **Key Insight:** Dynamic balance between texture (CNN) and structure (SSM) is crucial

Future Work:

- Extend to multi-speaker and zero-shot TTS
- Analyze learned gating weights α_i across layers
- Optimize for edge deployment (model compression, quantization)

Thank You!

Questions?

Code & audio samples: `https://github.com/huseyin-karaca/speechbrain`